

EEE 306 (July 2023)

Power System I Laboratory

Final Project Report

Section: C1 Group: 05

Droop Control in Decentralized Inverter-based AC, Islanded & DC Microgrid

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1 Abstract

Microgrids are emerging as a promising solution to integrate renewable energy sources, improve grid resilience, and enhance energy efficiency. Efficient control strategies are crucial for the stable operation of microgrids. Droop control is a widely adopted method for regulating voltage and frequency in microgrids, offering decentralized control and robustness against disturbances. We've presented the implementation of droop control in inverter-based AC, islanded, and DC microgrids using Simulink. The study focuses on understanding the behaviour of droop control under various operating conditions and its impact on system stability and performance. We begin with an overview of microgrid architecture and droop control principles. It then discusses the design and simulation setup in Simulink for inverter-based AC, islanded, and DC microgrids. Key components such as inverters, renewable energy sources, loads, and energy storage systems are modelled, and droop control algorithms are implemented for voltage and frequency regulation. Simulations are conducted to analyse the response of the microgrids under different scenarios, including variations in load demand, changes in renewable energy generation, and grid disturbances. The performance metrics such as voltage and frequency deviations, power sharing among sources, and system stability are evaluated and compared. Discussion on the effectiveness of droop control in maintaining stable operation and improving the overall performance of microgrids is proposed later. Insights gained from the simulations provide valuable guidance for the design and implementation of droop control strategies in real-world microgrid applications, contributing to the advancement of sustainable and resilient energy systems.

2 Introduction

Droop control is a widely adopted decentralized control strategy in microgrids, offering a simple yet effective method for regulating voltage and frequency. Originally developed for mechanical governors in diesel generators, droop control has been adapted for electronic inverters in modern microgrids. It provides a mechanism for power sharing among multiple sources while maintaining stable voltage and frequency levels, even in islanded operation when disconnected from the main grid. This project focuses on the implementation and analysis of droop control in three different types of microgrids: inverter-based AC microgrid, islanded microgrid, and DC microgrid. The use of Simulink, a powerful simulation tool, facilitates the modeling and simulation of these microgrid configurations, allowing for a comprehensive study of droop control performance under various operating conditions.

In this project, we aim to: Understand the fundamental principles of droop control and its application in different microgrid architectures. Model and simulate inverter-based AC, islanded, and DC microgrids using Simulink, incorporating various DERs and loads. Implement droop control algorithms for voltage and frequency regulation in each microgrid configuration. Analyze the performance of droop control under different scenarios, including changes in load demand, variations in renewable energy generation, and grid disturbances. Evaluate the effectiveness of droop control in maintaining system stability, optimizing power sharing, and enhancing overall microgrid performance.

By investigating droop control in these diverse microgrid setups, we aim to gain insights into its capabilities, limitations, and adaptability across different grid configurations. The findings of this project can inform the design and deployment of droop control strategies in real-world microgrid applications, contributing to the advancement of sustainable and resilient energy systems.

3 Design

3.1 Problem Formulation (PO(b))

3.1.1 Identification of Scope

Voltage Regulation: Droop control primarily regulates voltage within the microgrid. It identifies the voltage range within which droop control can effectively maintain stability. This includes determining the minimum and maximum acceptable voltage levels under different operating conditions.

Frequency Regulation: It also helps in regulating the frequency of the microgrid determining the acceptable frequency range under various loads and grid conditions. It can maintain frequency stability during load changes or disturbances.

System Stability: Droop control influences the overall stability of the system during transient events, such as sudden load changes or connection/disconnection of distributed energy resources (DERs).

Grid Integration: It interacts with the main grid during grid-connected operation and ensures seamless transition between grid-connected and islanded modes.

Load Sharing: Droop control facilitates load sharing among distributed energy resources ensuring fair and proportional sharing of loads among different DERs within the microgrid.

Dynamic Response: Evaluating the dynamic response of droop control under various operating conditions, it's found out that it's effectiveness in the response time and restoring voltage and frequency to their nominal values following disturbances is satisfactory.

3.1.2 Literature Review

-H. Wu, H. Sun, L. Cai, X. Tao, Simulation on control strategies of grid-connected inverters. in The 2nd International Symposium on Power Electronics for Distributed Generation Systems (IEEE, 2010), pp. 904–908

- Discusses the basic operation of inverter in power control strategies

-J.W. Kim, H.S. Choi, B.H. Cho, A novel droop method for converter parallel operation. IEEE Trans. Power Electron. 17(1), 25–32 (2002)

-Explores the method of droop control in parallel inverter operation

-S. Dahale, A. Das, N.M. Pindoriya, S. Rajendran, An overview of DC-DC converter topologies and controls in DC microgrid, in 2017 7th International Conference on Power Systems (ICPS). (IEEE, 2017, December), pp. 410–415

-Examines the impact of control on converters in DC microgrid

-A.K. Sahoo, K. Mahmud, M. Crittenden, J. Ravishankar, S. Padmanaban, F. Blaabjerg, Communication-less primary and secondary control in inverter-interfaced AC microgrid: an overview. IEEE J. Emerg. Select Top. Power Electron. (2020)
-Investigates the control of AC microgrid

-S. Luo, Z. Ye, R.L. Lin, F.C. Lee, A classification and evaluation of paralleling methods for power supply modules, in 30th Annual IEEE Power Electronics Specialists Conference Record (Cat. No. 99CH36321), vol. 2 (IEEE, 1999, July), pp. 901–908
-A comprehensive textbook that covers the evaluation of paralleling method.

3.1.3 Formulation of Problem

Voltage and Frequency Regulation:

- Voltage and frequency regulation are critical for maintaining the stability and reliability of the microgrid system.
- In AC microgrids, both voltage and frequency deviations must be kept within acceptable limits to ensure compatibility with connected loads and grid interconnections.
- In DC microgrids, voltage regulation is paramount due to the absence of natural frequency control mechanisms. Stable voltage levels are necessary to prevent issues such as voltage sag or voltage rise, which can impact the performance of connected loads.
- In islanded microgrids, where the system operates independently, maintaining nominal voltage and frequency levels is crucial for stable and reliable operation.

Power Sharing and Coordination:

- Effective power sharing among different sources (such as renewable energy systems, energy storage, and conventional generators) and loads is essential to optimize system efficiency and resource utilization.
- Droop control enables distributed energy resources (DERs) to autonomously adjust their output power based on deviations in voltage and frequency. However, coordinating the droop settings of multiple DERs is necessary to ensure proper power sharing and system stability.
- It includes determining appropriate droop slopes and reference values for each DER to achieve desired power sharing and voltage/frequency regulation objectives.

Stability and Resilience:

- Ensuring stability and resilience against disturbances, such as sudden load changes, fluctuations in renewable energy generation, or faults in the system, is a key consideration.
- It involves designing droop control algorithms that can respond effectively to disturbances while maintaining system stability and minimizing transient deviations in voltage and frequency.
- Advanced control strategies, such as adaptive droop control or hierarchical control schemes, may be employed to enhance the robustness and resilience of the microgrid system.

Scalability and Flexibility:

- It should address the scalability and flexibility requirements of the microgrid system, allowing for seamless integration of new DERs or changes in system configuration.

- Droop control algorithms should be adaptable to varying system conditions and capable of accommodating different types of DERs with diverse characteristics.

3.1.4 Analysis

AC Microgrid with Inverters:

In an AC microgrid with inverters, droop control is used to regulate both frequency and voltage. Each inverter is typically equipped with a droop control algorithm that adjusts its output power in response to deviations in grid frequency and voltage. The droop characteristic determines how much the output power of the inverter changes with variations in frequency and voltage. By adjusting the droop slope, the sensitivity of the inverter to frequency and voltage changes can be controlled. This enables effective power sharing among multiple inverters and ensures stability in the microgrid.

DC Microgrid with Inverters:

In a DC microgrid with inverters, droop control is used for voltage regulation. Since DC microgrids lack a natural mechanism for frequency control, droop control primarily focuses on maintaining voltage levels. Each inverter in the DC microgrid implements a droop control scheme that adjusts its output voltage based on deviations from the nominal voltage level. The droop characteristic determines the relationship between output voltage and load current. By adjusting the droop slope, the inverter's sensitivity to changes in load current can be controlled, ensuring proper voltage regulation and power sharing among multiple sources and loads.

Islanded Microgrid:

In an islanded microgrid, where the system operates independently from the main grid, droop control is crucial for both frequency and voltage regulation. Each distributed energy resource (DER) within the microgrid, such as renewable energy sources and energy storage systems, employs droop control to adjust its output power in response to deviations in frequency and voltage. The droop characteristics of the DERs determine their contribution to frequency and voltage regulation within the microgrid. Proper coordination of droop settings ensures stable operation and effective power sharing among various DERs, enabling the islanded microgrid to maintain reliable and resilient power supply.

3.2 Design Method (PO(a))

AC:

A. Droop control: Droop control is not a novel concept. It is generally used in the parallel operation of synchronous machines. It is also employed for the parallel functioning of inverters for accurate and proportional load sharing. In this method, three phase voltage and current are converted into a direct-quadrature-zero reference frame [1] which reduces the three variable quantity to two variable and AC to DC. The below given block diagram fig.2 is the control scheme of the inverter for parallel operation. Three controllers generate the pulse width modulation for the inverter (voltage, current and power) as shown in fig.2.

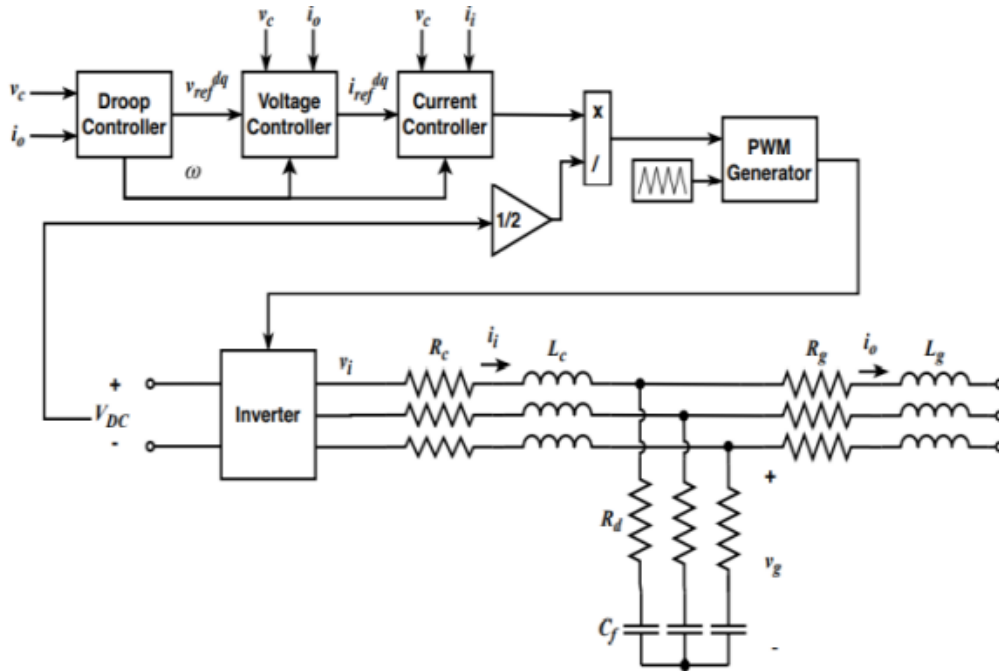


Fig. 2. Block diagram of parallel operated inverter using droop control

B. Controllers – The voltage controller regulates the voltage across the capacitor as well as the damping resistance. To generate reference currents and ref voltages, the droop control block creates ref voltage, which is passed into the back-to-back voltage controller and current controller respectively ($i^* d$ and $i^* q$) and ($v^* d$ and $v^* q$).

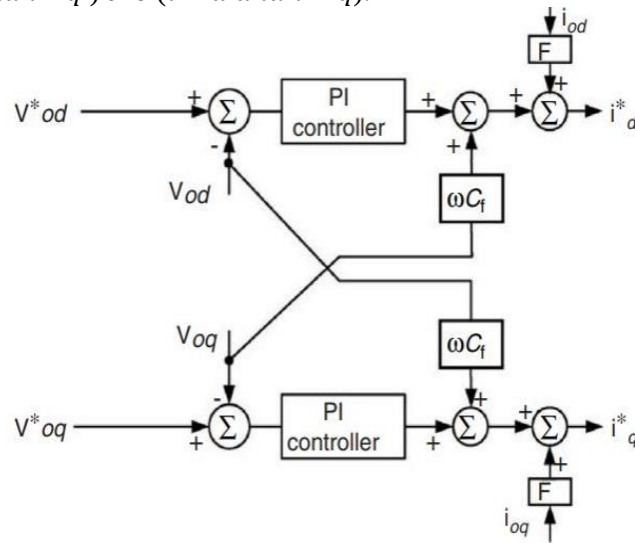


Fig. 3. Voltage controller block diagram

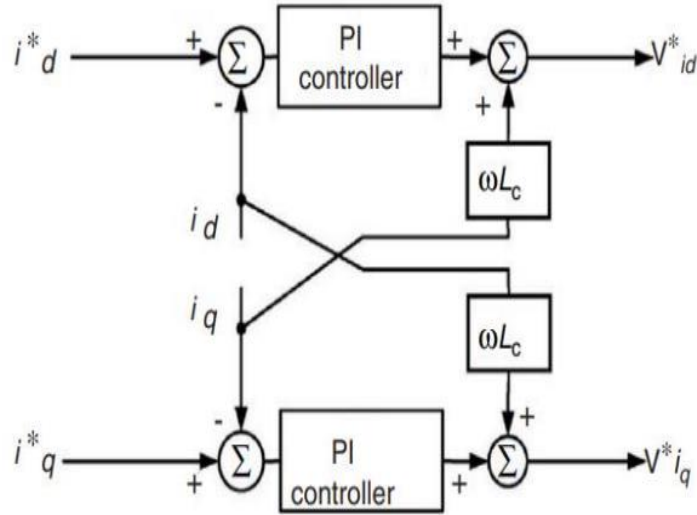


Fig. 4. Current controller block diagram

Both controllers as shown in the Fig.3 and Fig.4, regulates the voltage and current. To provide decoupling among controllers, the current controller should have a faster reaction than the voltage controller [9]-[10].

The voltage and current controller is passed through P-I controller, which is tuned by pole placement [9]. When Kirchhoff's current law is applied to the single phase equivalent of the LCL filter in fig. 2, the outcome is

$$i_L \approx I_{out} + Cf \, dV_c / dt \quad (1)$$

For this single phase equivalent in the -frame, the equation remains the same and utilizing the relationship between d-q

$$Cf \, dV_c^d / dt - \omega Cf V_c^d = i_L^d - I_{out}^d \quad (2)$$

$$Cf \, dV_c^q / dt + \omega Cf V_c^d = i_L^q - I_{out}^q \quad (3)$$

When we transform equations (2) and (3) into the frequency domain, we obtain

$$V_c^d = (i_L^d - I_{out}^d + \omega Cf V_c^d) (1 / sCf) \quad (4)$$

$$V_c^q = (i_L^q - I_{out}^q - \omega Cf V_c^q) (1 / sCf) \quad (5)$$

By using P-I controller, output of the voltage controller generates the output of the current controllers.

$$i_{ref}^d = I_{out}^d + (V_{out,ref}^d - V_c^d) (K_{pv} + K_{iv} / s) - V_c^q \omega Cf \quad (6)$$

$$i_{ref}^q = I_{out}^q + (V_{out,ref}^q - V_c^q) (K_{pv} + K_{iv} / s) + V_c^d \omega Cf \quad (7)$$

By using KVL in fig.2 for current controller

$$V_i = i_L R_c + L_c \, di_L / dt + V_c \quad (8)$$

Changing the into d-q reference frame

$$L_c \, di_L^d / dt - \omega L_c i_L^q = V_i^d - R_c i_L^d - V_c^d \quad (9)$$

$$L_c \, di_L^q / dt + \omega L_c i_L^d = V_i^q - R_c i_L^q - V_c^q \quad (10)$$

Changing equation into frequency domain

$$i_L^d = (V_i^d - V_c^d + \omega L_c i_L^q) (1 / L_c) / (s + R_c / L_c) \quad (11)$$

$$i_L^q = (V_i^q - V_c^q - \omega L_c i_L^d) (1 / L_c) / (s + R_c / L_c) \quad (12)$$

The current controller provides ref voltage which is given below

$$v_{ref}^d = V_o^d + (i_{ref}^d - i_L^d) (K_{pc} + K_{ic} / s) - \omega L_c i_L^q \quad (13)$$

$$v_{ref}^q = V_o^q + (i_{ref}^q - i_L^q) (K_{pc} + K_{ic} / s) + \omega L_c i_L^d \quad (14)$$

C. Real and Reactive Power in Parallel operation

When two inverters are operating in parallel than their equivalent circuit diagram is shown below in the fig. 5.

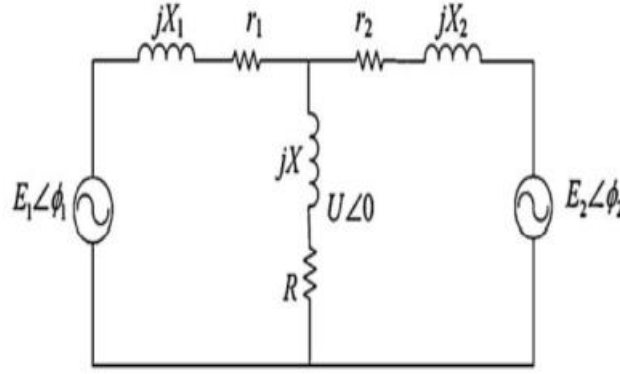


Fig. 5. Equivalent circuit diagram of two parallel operated inverter

Their output voltages are $E_k \angle \phi_k$ ($k=1, 2$), and ϕ_k is the voltage phase angle of each DG. The impedance of feeder is noted with $r_k + jX_k$ ($k=1, 2$) and load $R + jX$ with voltage $U \angle 0$. As a result, each inverter's active and reactive power is computed as follows [11].

$$P_k = (1/|Z_k|)^* ((U E_k \cos(\phi_k) - U^2) \cos(\theta_k) + U E_k \sin \phi_k \sin \theta_k) \quad (15)$$

$$Q_k = (1/|Z_k|)^* ((U E_k \cos(\phi_k) - U^2) - U E_k \sin \phi_k \cos \theta_k) \quad (16)$$

$$\text{Where } Z_k = r_k + jX_k \rightarrow \{|Z_k| = \sqrt{r_k^2 + X_k^2}\}$$

$$\theta_k = \tan^{-1}(X_k/r_k)$$

equations (15) and (16) can be simplified for different types of impedance values [8] as follows:

For resistive impedance, inductive part will be negligible $\theta_k = 0^\circ$

$$P_k = (1/|r_k|)^* ((U E_k \cos(\phi_k) - U^2)) \quad (17)$$

$$Q_k = - (1/|Z_k|)^* (U E_k \sin \phi_k) \quad (18)$$

So, resistive droop control will become

$$E_k = E_k^* - m_k P_k \quad (19)$$

$$w_k = w_k^* + n_k Q_k \quad (20)$$

For inductive droop control

$$\theta_k = 90^\circ \quad E_k = E_k^* - m_k Q_k \quad (21)$$

$$w_k = w_k^* - n_k P_k \quad (22)$$

Islanded:

The main objective of microgrids in islanded mode is to allow the system to operate even in adverse scenarios, such as faults in the main grid, high prices of main grid's power and supplying in remote areas.

Primary Control:

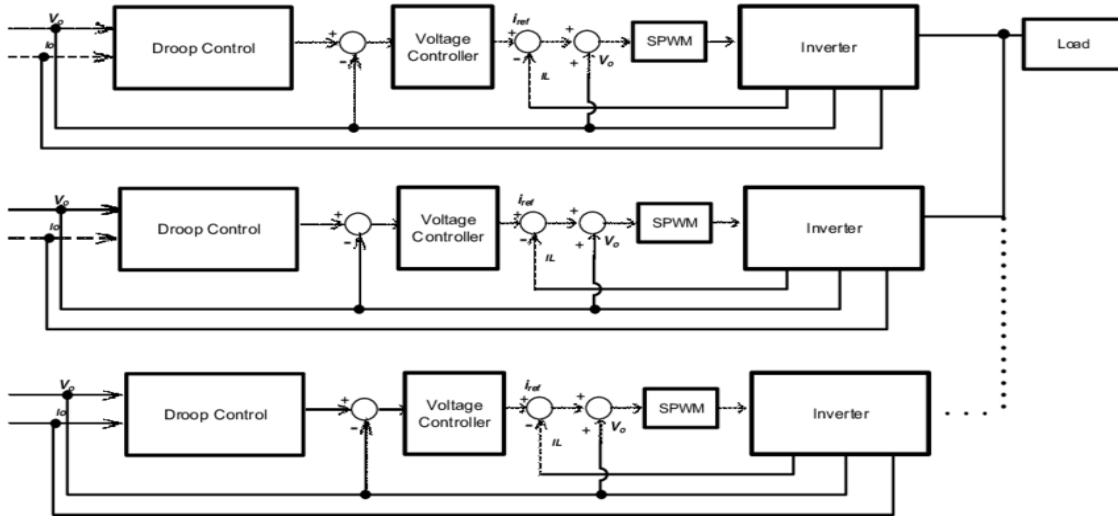
Droop method is used to emulate the behavior of synchronous machines. In this level, the virtual impedance can also be implemented to improve X/R ratio.

Islanded Microgrid Operation:

A microgrid is said to be in islanded mode when it's disconnected from the main grid & it operated independently with micro sources and loads.

Islanded Mode refers to the case when a circuit breaker isolated the microgrid from the rest of the utility grid.

Block Diagram:



DC:

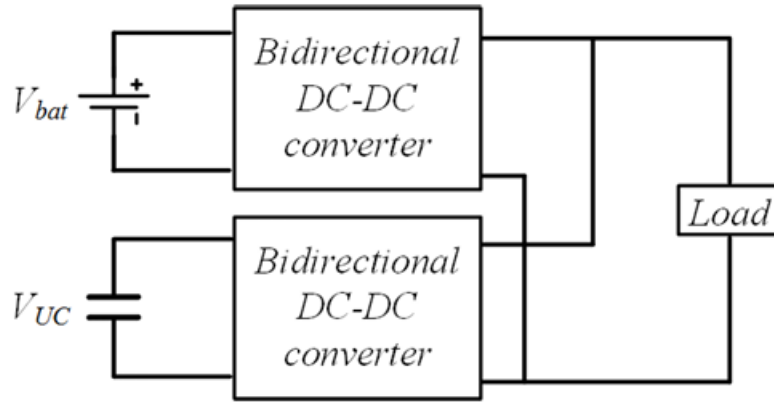


Fig: Block diagram of Droop Control of DC Microgrid circuit

FREQUENCY COORDINATING VIRTUAL IMPEDANCE CONCEPT:

The output voltage reference in droop control is given by

$$v = V_{ref} - iR_{droop}$$

Where, v and i are static terminal voltage and current of the converter respectively, and V_{ref} is the global voltage reference.

The term iR_{droop} characterizes the output impedance of the converter and is produced by the droop controller. However, this does not expedite transient power regulation since it only describes the static power sharing among individual sources. Conversion of R_{droop} to Z_{droop} provides effective frequency domain shaping of the virtual output impedance. Equation (3) signifies power scale parameters in the form of R_{droop} to determine steady state power flow and $F(s)$ determines transient behaviors.

$$v(s) = v_{ref}(s) - i(s)Z_{droop}(s)$$

$$Z_{droop}(s) = R_{droop}F(s)$$

The control loop of the converter in the DC microgrid with an implementation of virtual impedance is shown in Fig. 6 [3]. Reference current (i_{ref}) for the current control loop of the converter can be written as in (4) where $G_{conv}(s)$ represents the transfer function for inner closed loop current control, v_{dc} and i_{dc} represent the voltage across and current injected in the DC network, respectively. The converter output impedance (Z_{conv}) is expressed as in (5).

$$i_{ref}(s) = \frac{v_{ref}(s) - v_{dc}(s)}{Z_{droop}(s)}$$

$$Z_{conv}(s) = \frac{v_{ref}(s) - v_{dc}(s)}{i_{dc}(s)} = \frac{Z_{droop}}{G_{conv}(s)}$$

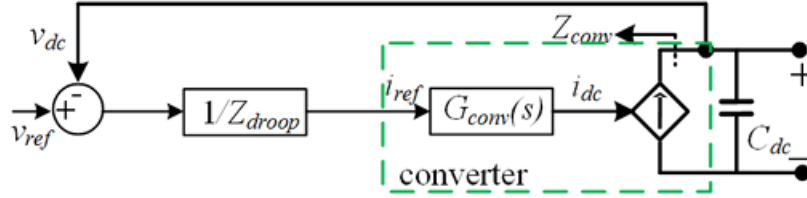


Fig . Control loop for virtual impedance concept with converter

CONTROL STRATEGY

The control loops for battery and SC converters, with series low pass filter (LPF) and high pass filter (HPF) respectively to highlight the frequency response are presented in Fig 7. The equivalent impedances for battery (Z_{bat}) and SC (Z_{sup}) converters using first order LPF and HPF are given by (6) and (7), respectively. The equivalent impedance (Z_{ESS}) for the entire HESS is given by (8).

$$Z_{bat}(s) = \frac{R_{droop} (\omega_c + s)}{G_{conv}(s) \omega_c}$$

$$Z_{sup}(s) = \frac{R_{droop} (\omega_c + s)}{G_{conv}(s) s}$$

$$Z_{ESS} = Z_{bat}(s) \parallel Z_{sup}(s) = \frac{R_{droop}}{G_{conv}(s)}$$

For better attenuation of ripple power, the above control loop can be modified by inserting higher order filters. A higher order filter will attenuate frequencies outside its passband faster than a lower order filter. The modified equations using second order LPF and HPF for the control loops in Fig 7 are given by (9) and (10). The impedance versus frequency plots are shown in Fig. 8 and Fig. 9 for the control loops with first order and second order filters respectively. Z_{bat} and Z_{sup} intersect at the cut-off frequency (ω_c). The two frequencies ω_c and converter frequency ω_{conv} explain the response time scales for battery and SC. At frequencies below ω_c , battery supplies energy since Z_{ESS} is dominated by battery, while at higher frequency range between ω_c and ω_{conv} , SC responds to the power fluctuations. Hence, the complementary benefits of the battery and the SC in the hybrid energy storage is realized. The change in the roll off rate [Z (dB)] in Fig. 9 proves that the use of a second order filter provides better filtering of ripple power.

$$Z'_{bat}(s) = \frac{R_{droop} (\omega_c^2 + 2\xi\omega_c s + s^2)}{G_{conv}(s) \omega_c^2}$$

$$Z'_{sup}(s) = \frac{R_{droop} (\omega_c^2 + 2\xi\omega_c s + s^2)}{G_{conv}(s) s^2}$$

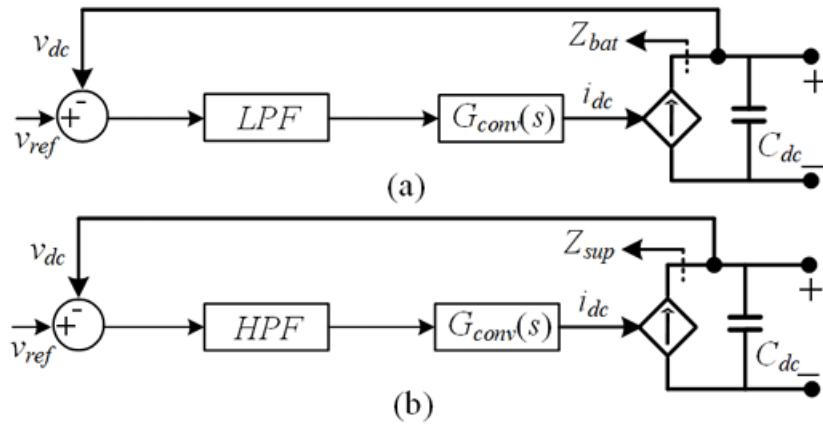


Fig : Control loops a) battery converter b) Supercapacitor converter.

B. Supercapacitor Voltage Restoration

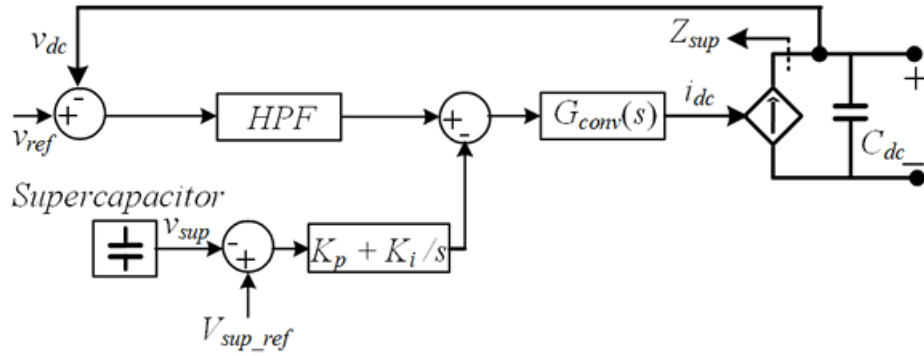


Fig : Control loop for Supercapacitor voltage restoration.

Component List:

- PV Array
- DC-DC Boost Converter
- Bi Directional DC-DC Converter Super Capacitor
- Combined Control Battery
- Supercapacitor
- RLC Load

3.3 Circuit Diagram for AC Microgrid:

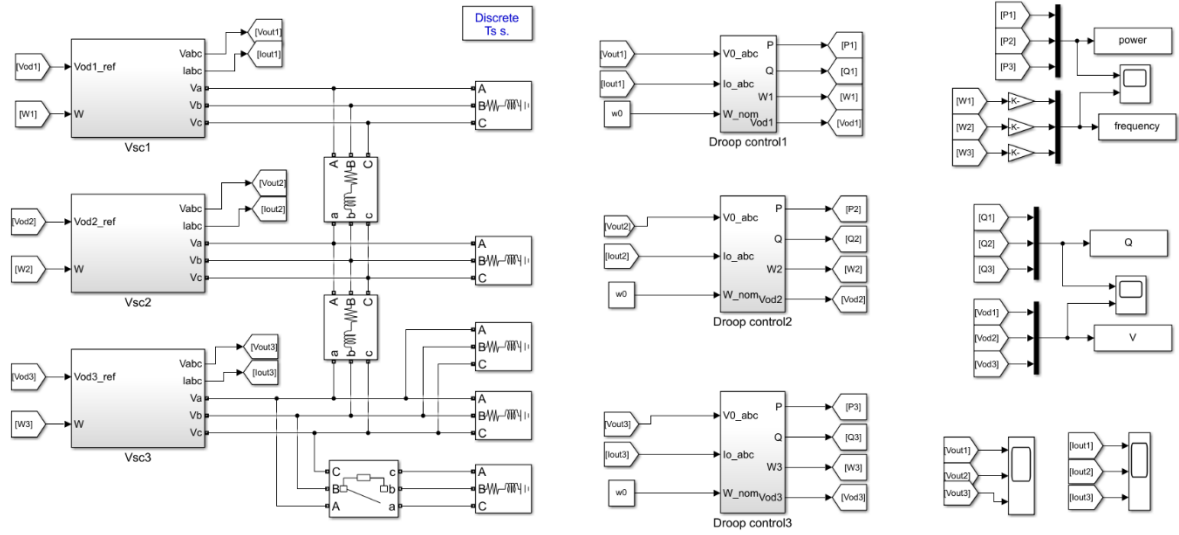


Fig: Droop Control in Inverter Based AC microgrid

In this simulation, microgrid consists of three VSCs which are connected to different loads. Each VSC consists of a droop controller along with outer voltage controller and inner current controller.

Droop originates from the principle of power balance in synchronous generators. An imbalance between the input mechanical power and the output electric power causes a change in the rotor speed and electrical frequency. Similarly, variation in output reactive power results in voltage magnitude deviation.

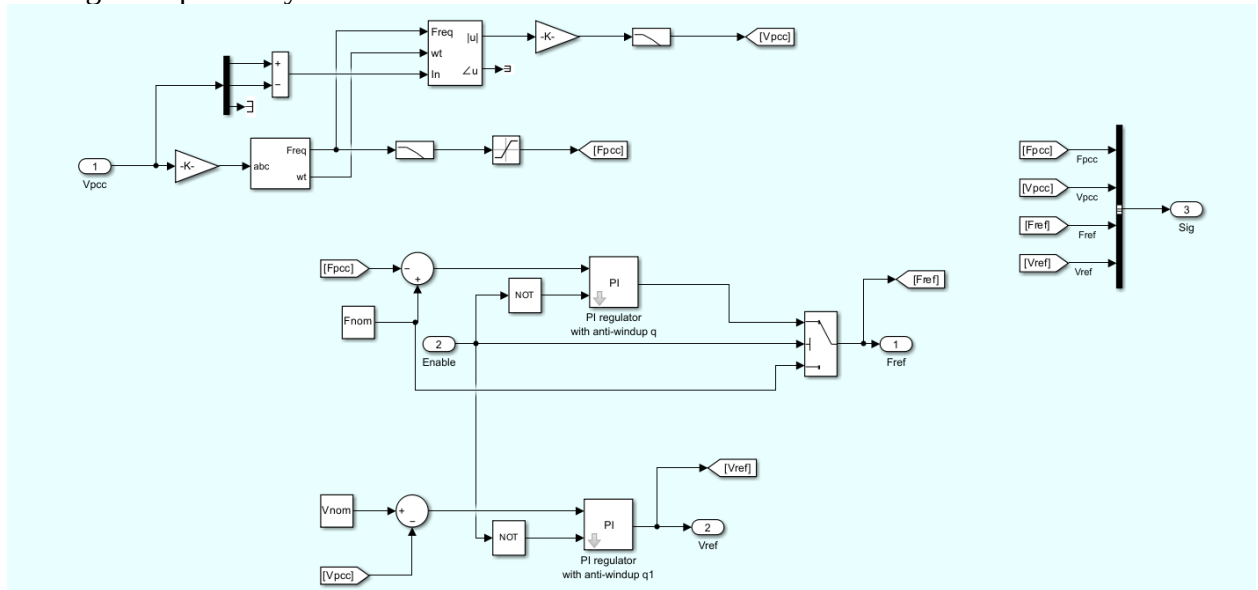
Such a characteristic can be artificially created for electronically interfaced inverter-based AC microgrid. In droop control, the relationships between real power and frequency and reactive power and voltage are as follows:

$$\omega_{ref} = \omega_{nominal} - m_p * P$$

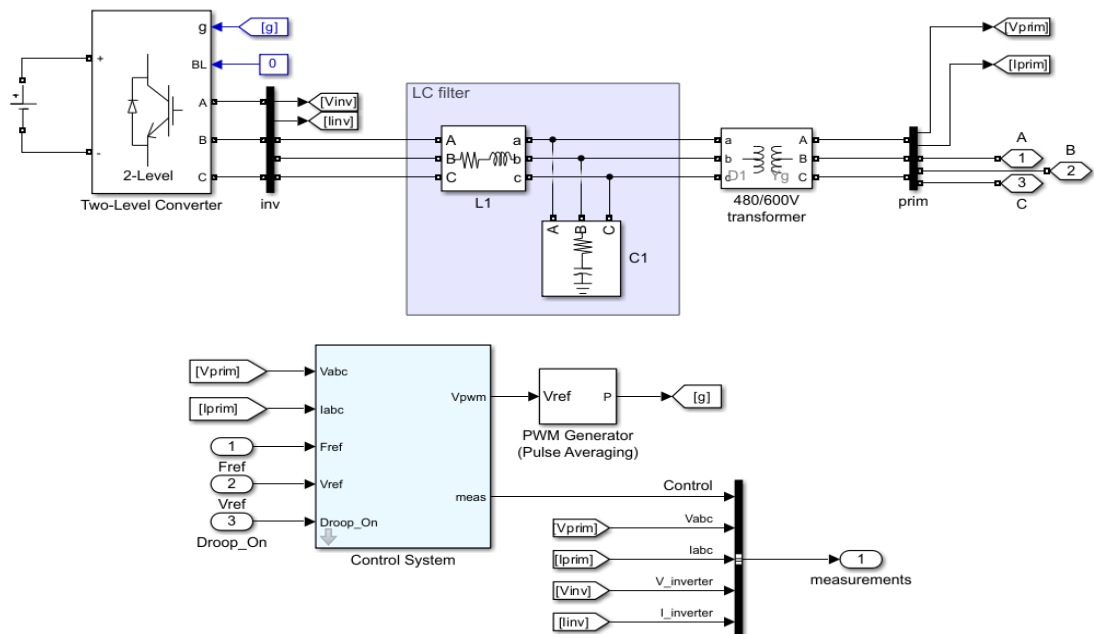
$$v_{ref} = v_{nominal} - n_q * Q$$

Circuit Diagram for Islanded Microgrid:

Microgrid Supervisory Control:



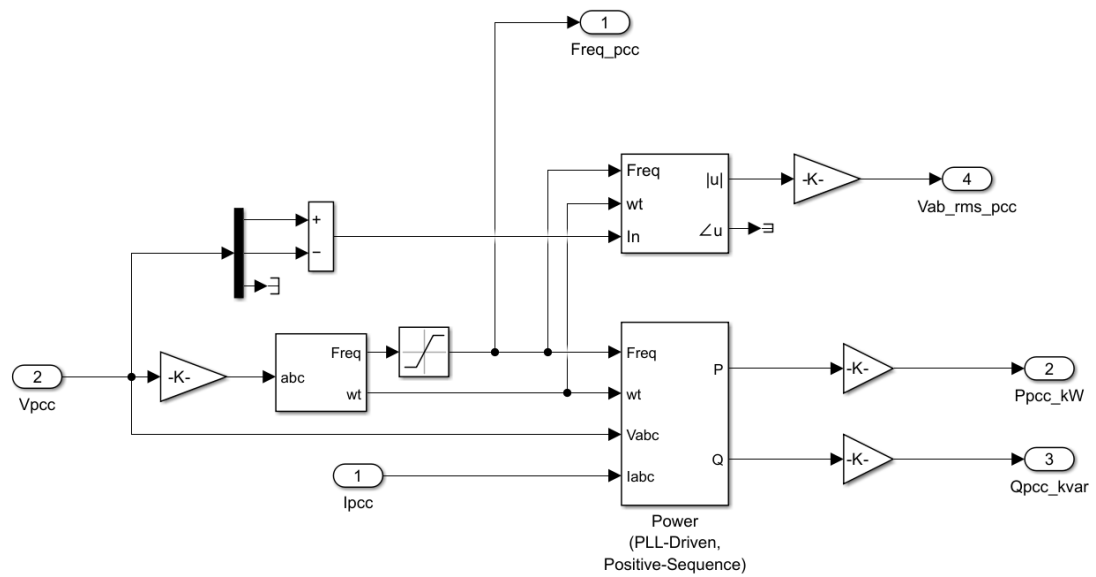
Inverter:



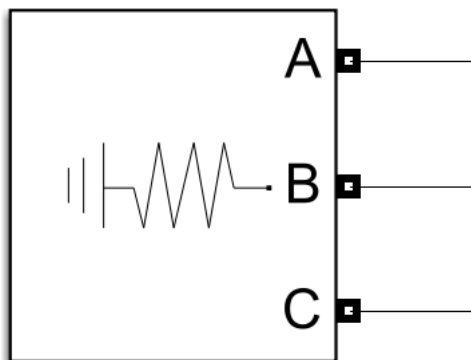
Each inverter subsystem contains a three-phase two-level power converter, an LC filter, a 480/600V transformer as well as an ideal DC source to represent the DC link of a typical renewable energy generation system (such as PV array, wind turbine, battery energy storage system).

Each subsystem also includes a control system and a PWM generator feeding the inverter.

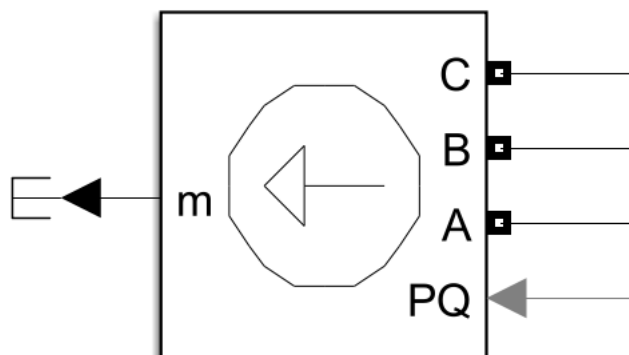
PCC Measurements:



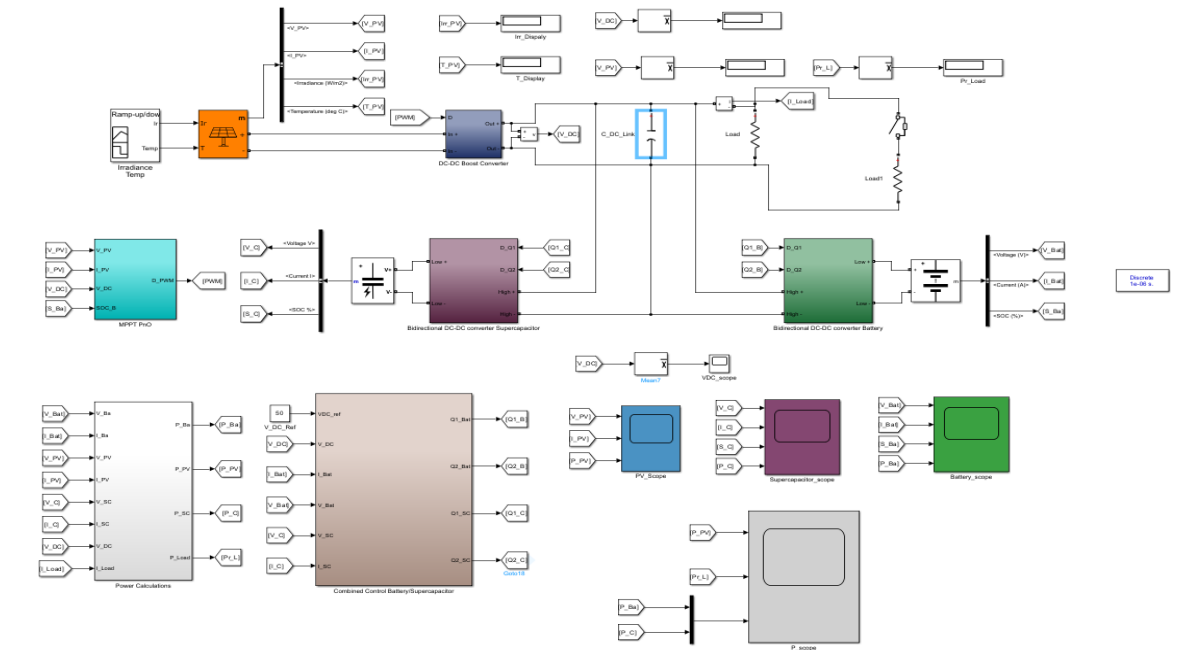
Static Load:



Variable Load:

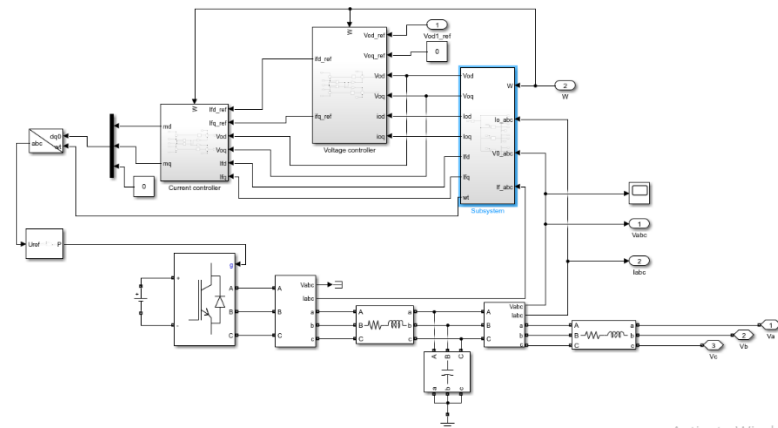


Circuit Diagram for DC Microgrid:



3.4 Simulation Model

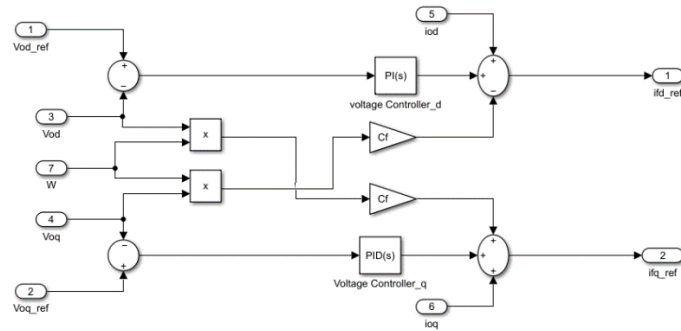
Voltage Source Converter Block:



Here we can see the the internal structure of the block. A LCL filter is used after inverter. The parameters of LCL filter is:

System Parameter	LCL Filter Parameter
Phase to Phase rms Voltage $V_n=380V$	$L_c=0.0035$, $R_c=0.1\text{ohm}$
F_g (Grid Frequency)= 50Hz	$L_g=0.00035$, $R_g=0.05$
Switching Frequency(f_{sw})= 10kHz	$C_f=5e-05$

Voltage Controller Block:



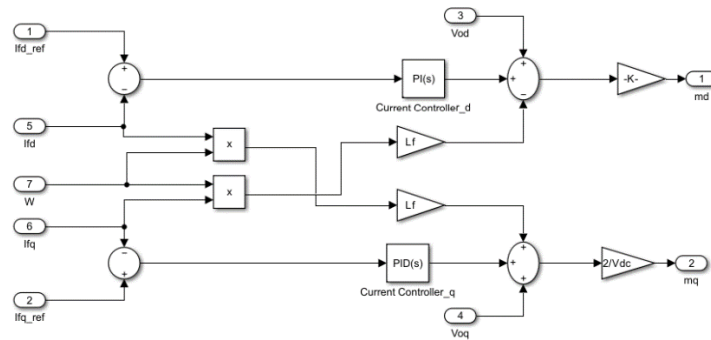
Here the parameters for PI Controller are,

$$K_{pv}=0.28586$$

$$K_{iv}=594.85$$

For the PID Controller, K_{pv} & K_{iv} remains the same, but the filter coefficient is chosen to be $N=100$. Subscript d & q means direct and quadrature axis component.

Current Controller Block:



Here the parameters of PI controller are,

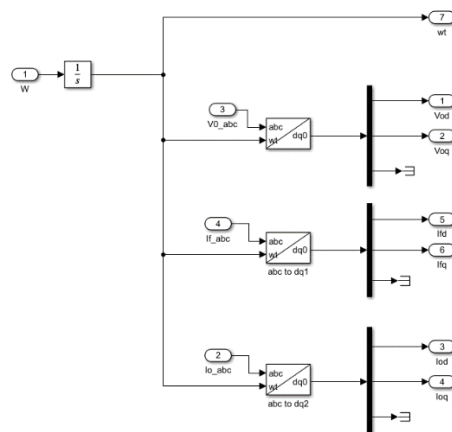
$$K_{pc}=54.978$$

$$K_{ic}=1570.8$$

For the PID controller, values of K_{pc} & K_{ic} remains the same, but the filter coefficient is chosen to be $N=100$ as before. The value of gain is,

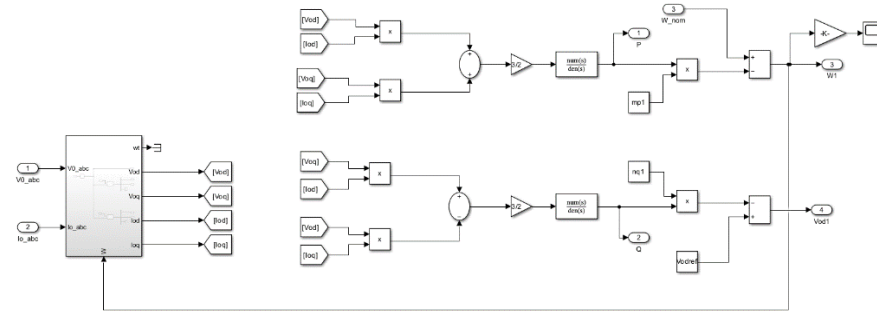
$$K=2/V_{dc}=0.0028571$$

Subsystem Block:



The block parameter for abc to dq0 transformation mask,
Rotating frame alignment at $wt=0$: 90° behind phase A axis

Droop Controller Block:



The numerator coefficient num(s)=[4*pi]

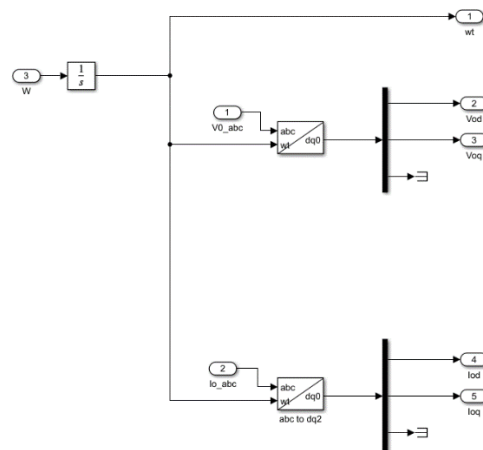
The denominator coefficient Den(s)=[1 4*pi]

Parameter	Symbol	Values
Active Droop Coefficients	mp1	5e-05
	mp2	
	mp3	
Reactive Droop Coefficients	nq1	0.003
	nq2	
	nq3	

Here, $w_{nom} = 2\pi \cdot f_{nom} = 2\pi \cdot 50 = 314$

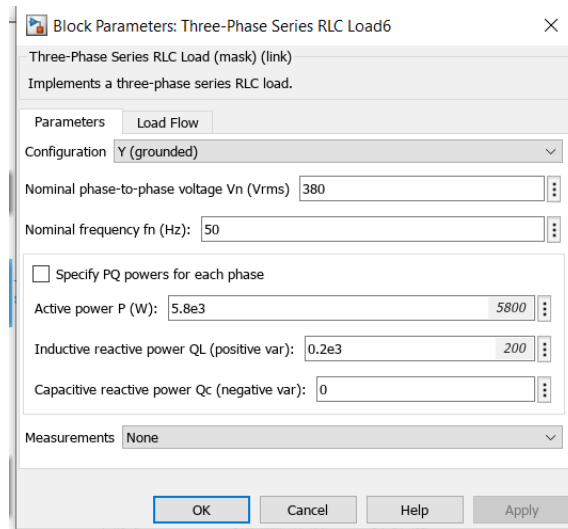
Gain $K = 1/2\pi = 0.15915$

Subsystem Block in the Droop Controller:

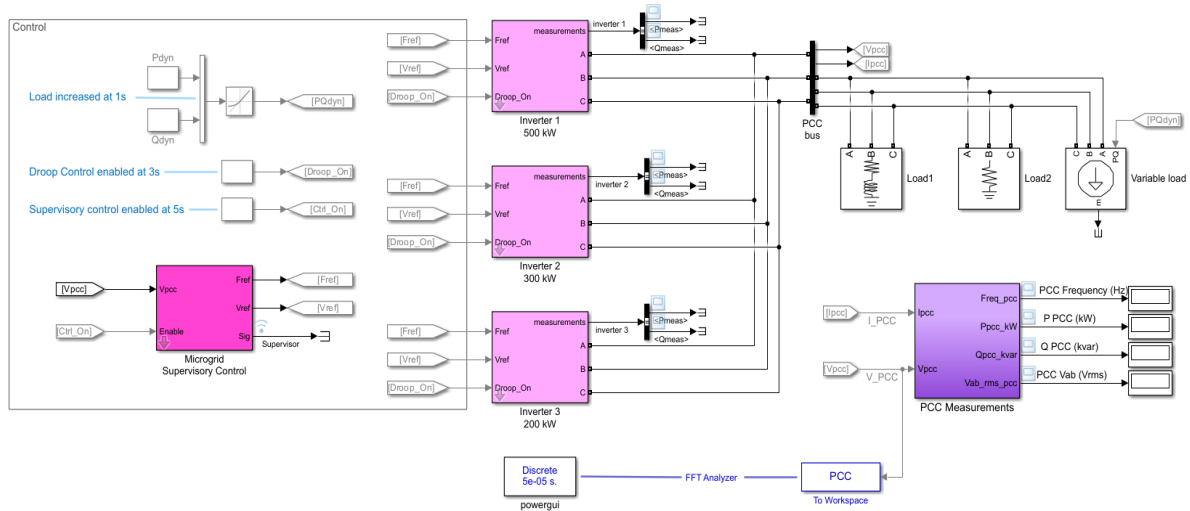


The rotating frame alignment for this is also 90 degrees behind phase A axis.

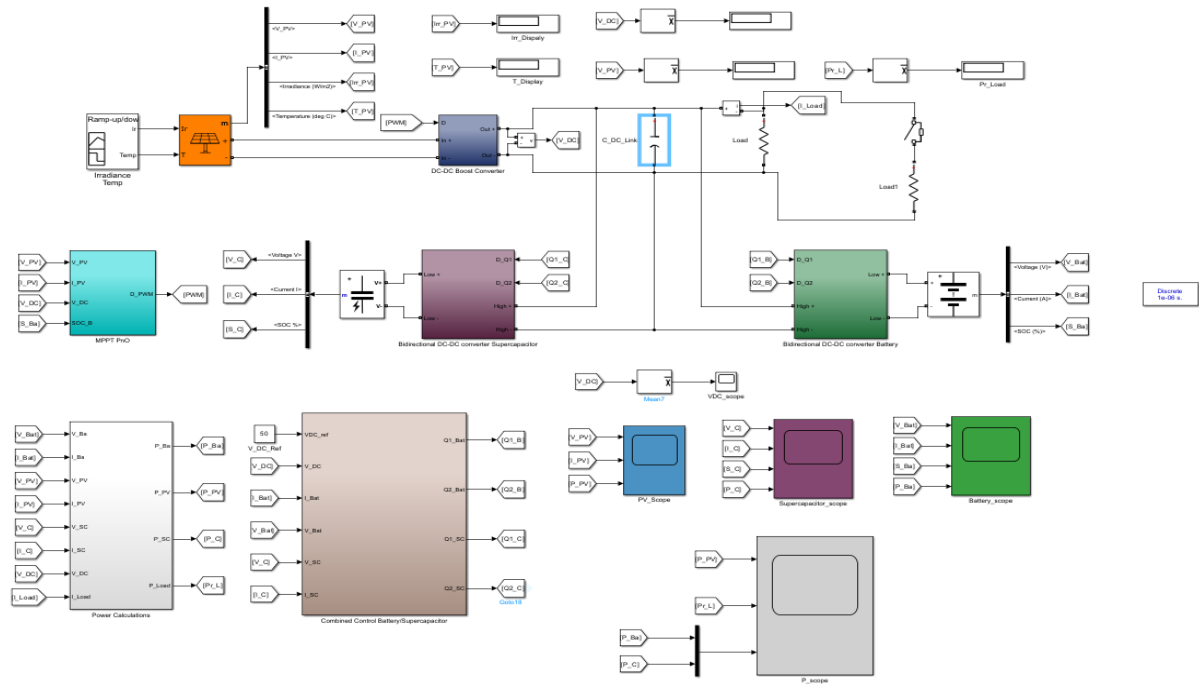
Three phase RLC Loads:



Islanded Simulation Model:



DC Simulation Model:



Photovoltaic (PV) arrays are integrated into droop control of DC microgrids primarily for their capability to provide renewable energy generation. This integration allows the microgrid to utilize solar power efficiently while maintaining voltage stability through droop control.

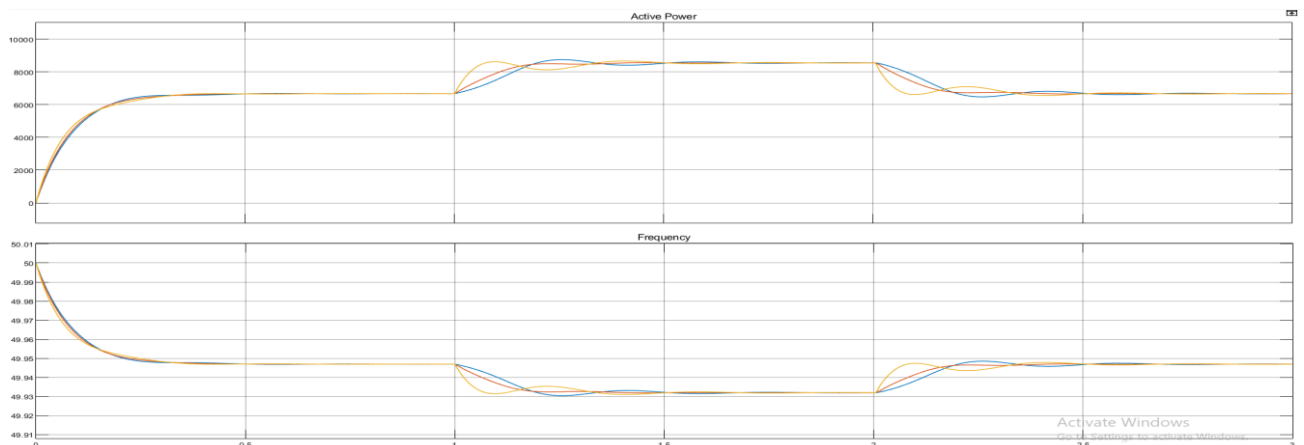
A DC-DC boost converter is used in droop control of DC microgrids for voltage regulation purposes. It adjusts the output voltage to match the desired reference voltage set by the droop control algorithm, ensuring stability despite load or generation fluctuations.

A bidirectional DC-DC converter with supercapacitors is used in droop control of DC microgrids mainly for its capability to efficiently manage energy flow bidirectionally and provide fast response times for voltage regulation. This setup allows for effective energy storage and discharge, contributing to the stability and reliability of the microgrid system.

4 Implementation

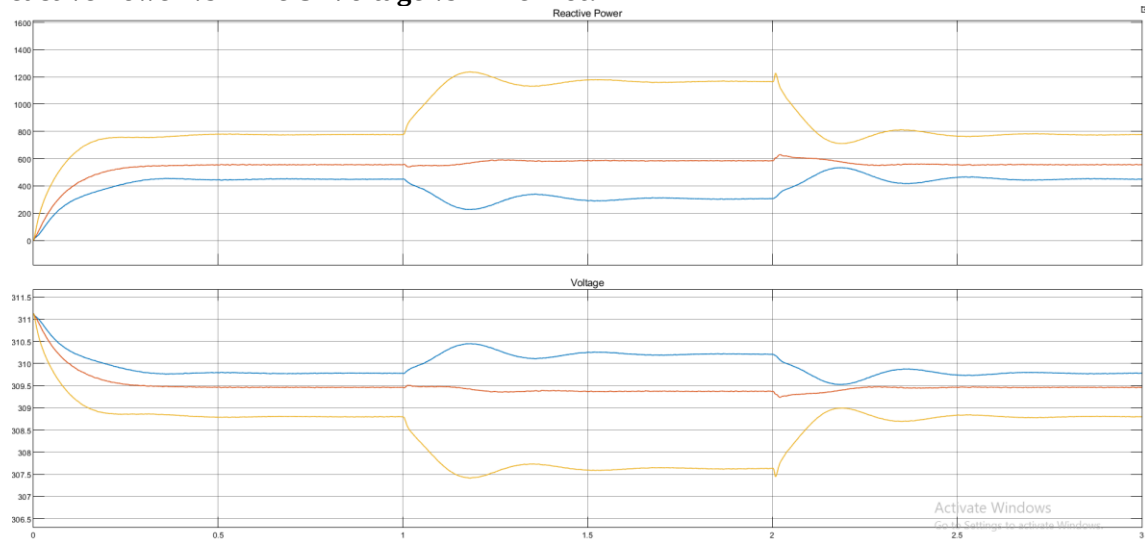
4.1 Description

Active Power vs Time & Frequency vs Time Plot:



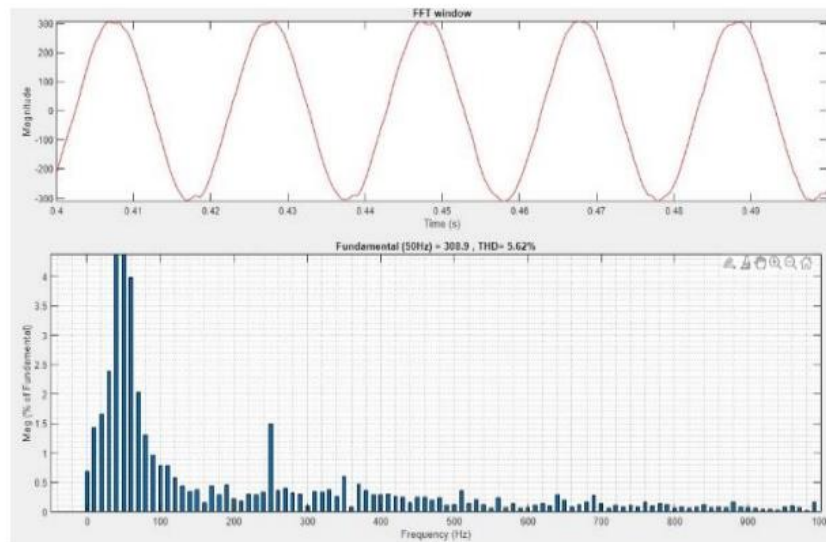
With the same feeder impedance and same droop coefficient, the load sharing between three inverters (active and reactive power) are same. As we increase the load at 1 sec to 2sec both inverters shares power (active and reactive) and voltage and frequency variations are same.

Reactive Power vs Time & Voltage vs Time Plot:

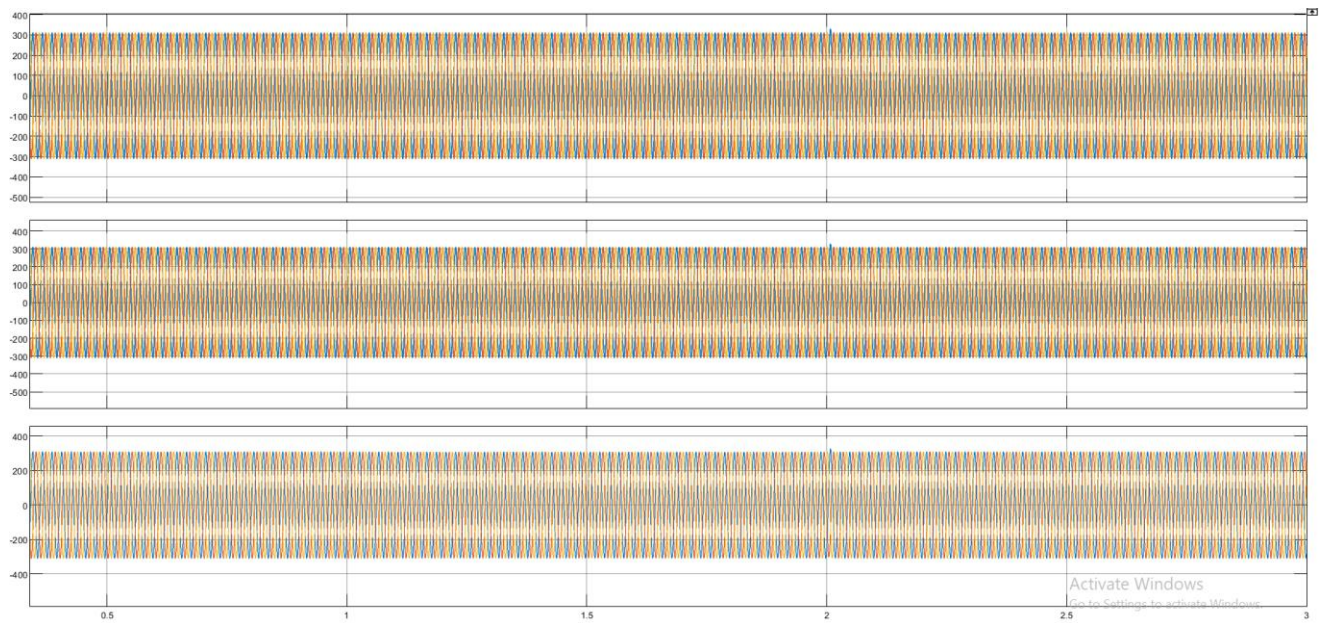


Here The switching Occurred between 1 and 2s, That's why the fluctuation was visible. After 2s when the controller activated, then power, frequency, reactive power, voltage all went to their nominal values eventually.

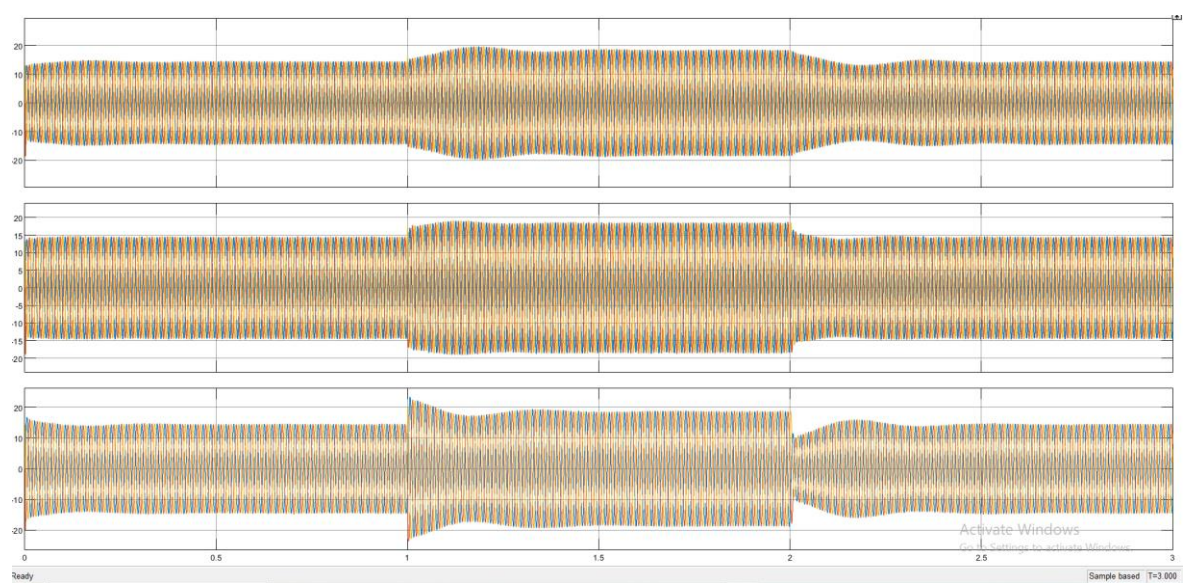
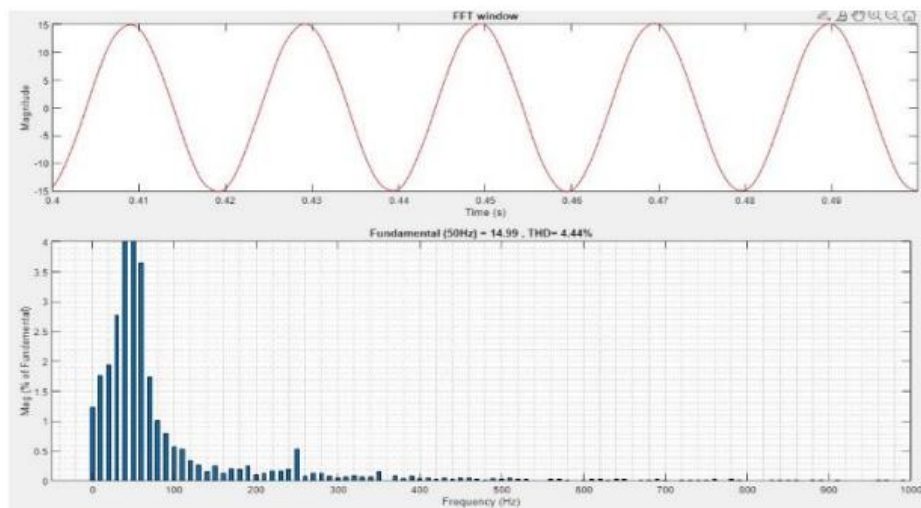
Output Voltage vs Time:



As the frequency is high, output voltage of only 0.5s is shown. But the output voltage remains the same in this whole 3s.

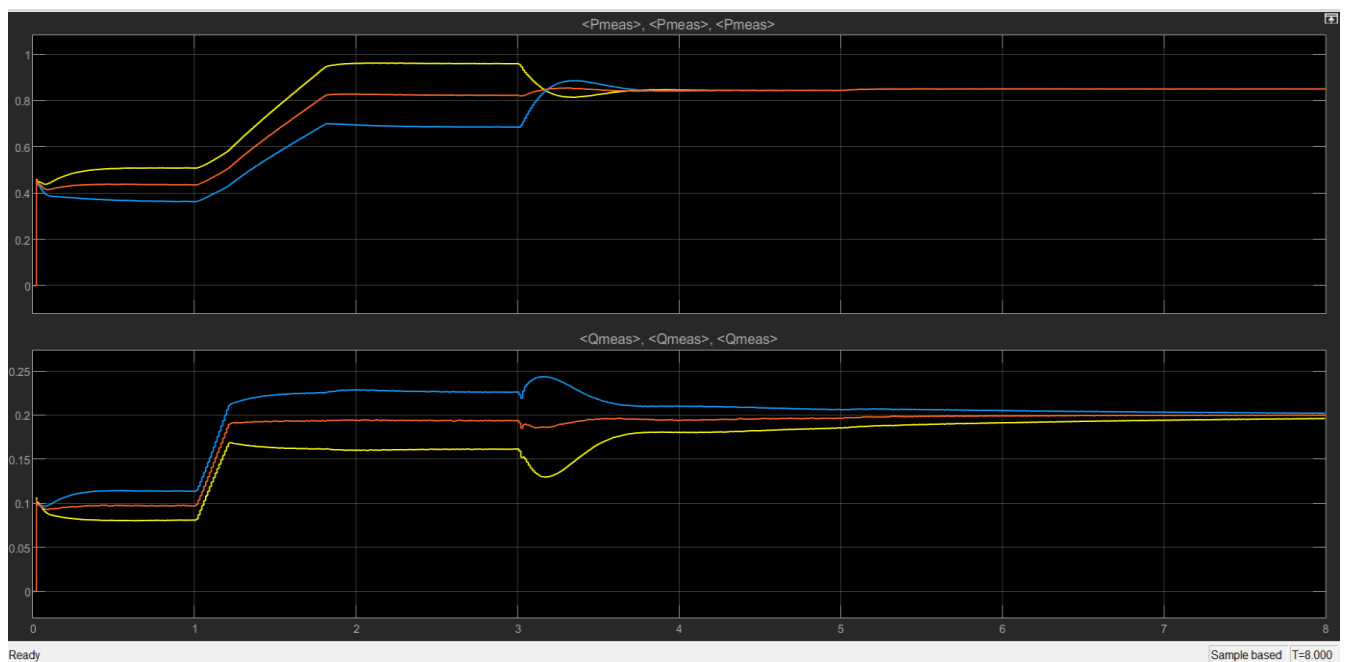
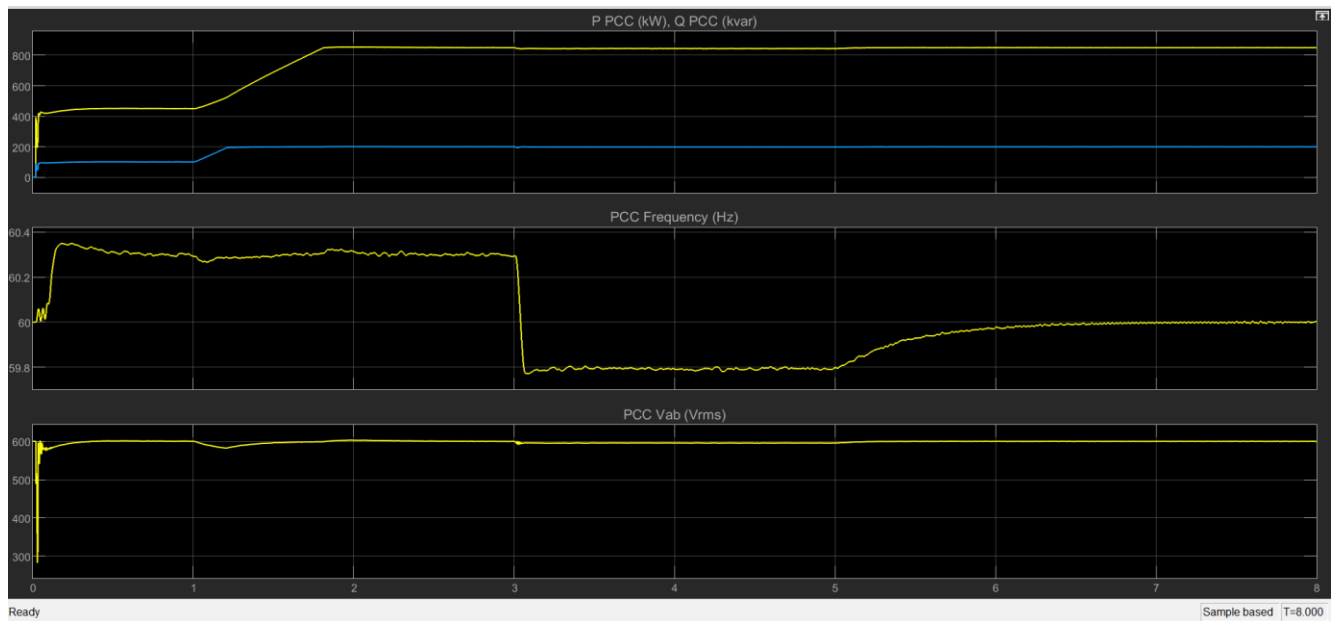


Output Current vs Time:



When load changes, some fluctuations are visible for output current but no fluctuation is seen in output voltage.

Islanded:



- The microgrid consists of three parallel inverter subsystems, with power ratings of 500 kW, 300 kW and 200 kW respectively.
- A dynamic load model is used to dynamically change the microgrid total load.
- Each subsystem also includes a control system and a SPWM generator feeding the inverter.
- All the inverter work at 60 Hz frequency.
- Line voltage 600 Vrms.

In droop control:

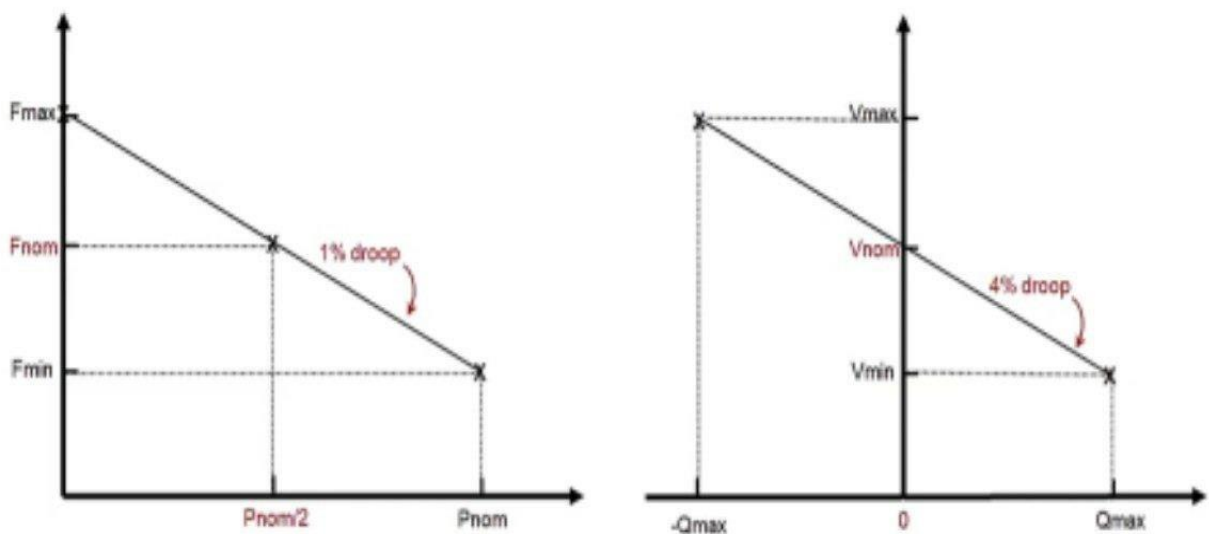
- Load increased at 1s

- Droop Control enabled at 3s
- Supervisory control enabled at 5s

The droop P/F is set to 1% meaning that microgrid frequency is allowed to vary from 60.3 Hz (inverter produces no active power) to 59.7 Hz (inverter produces nominal active power).

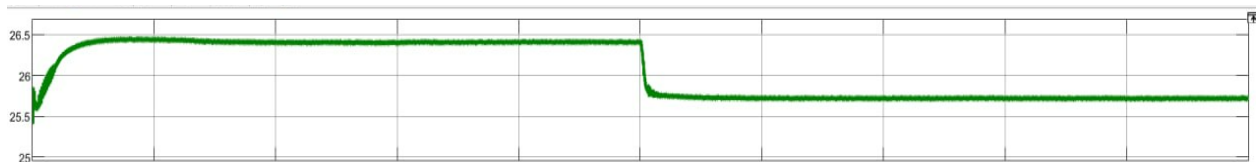
The Droop Q/V is set to 4% meaning that the microgrid voltage at the PCC bus is allowed to vary from 612 Vrms (inverter produces its full inductive power) to 588 Vrms (inverter produces full capacitive power).

Q_{max} is specified as half of the nominal active power P_{nom} .

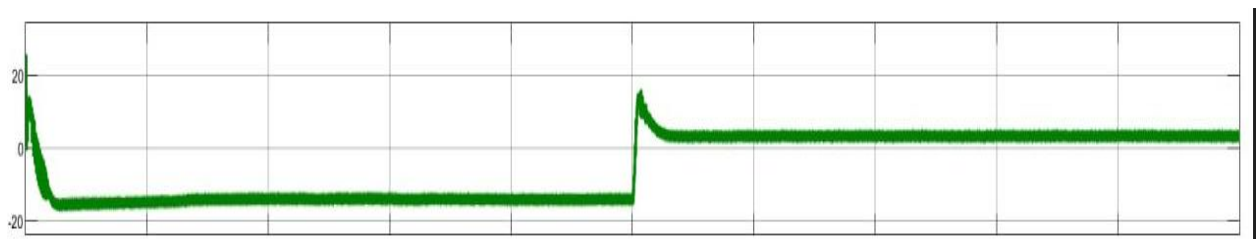


DC:

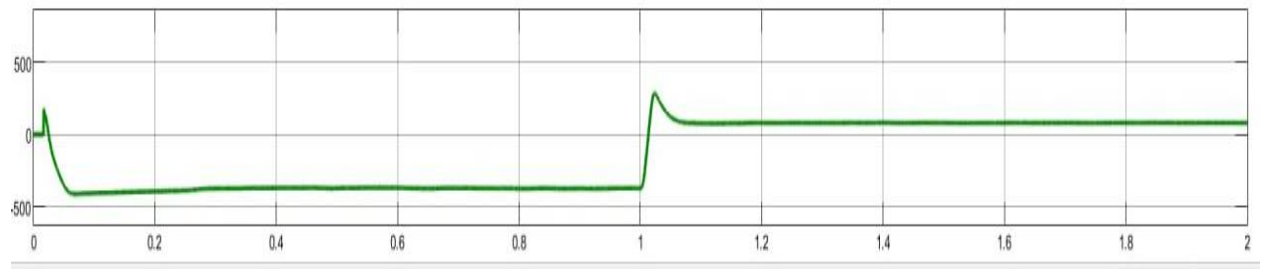
➤ Battery Voltage Vs Time



➤ Battery Current Vs time



Battery Power Vs time



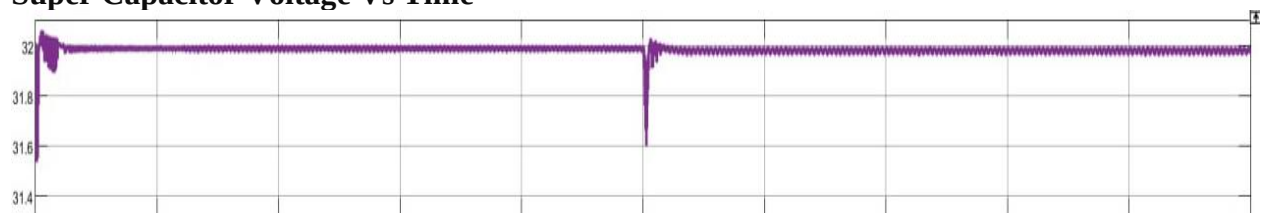
PV Power Vs Time



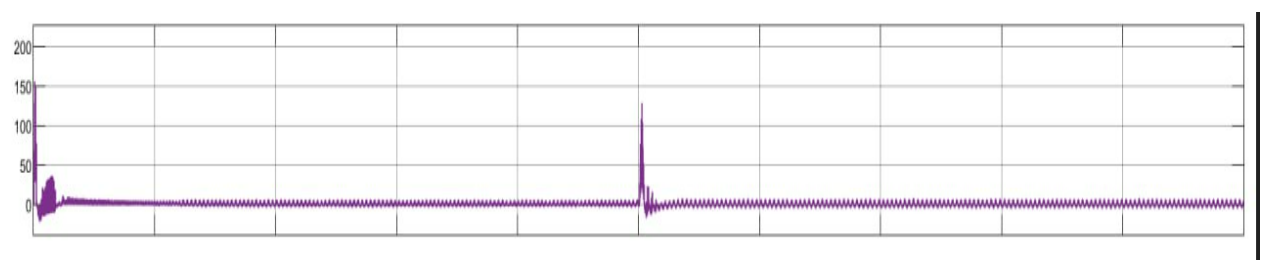
Load Power Vs Time



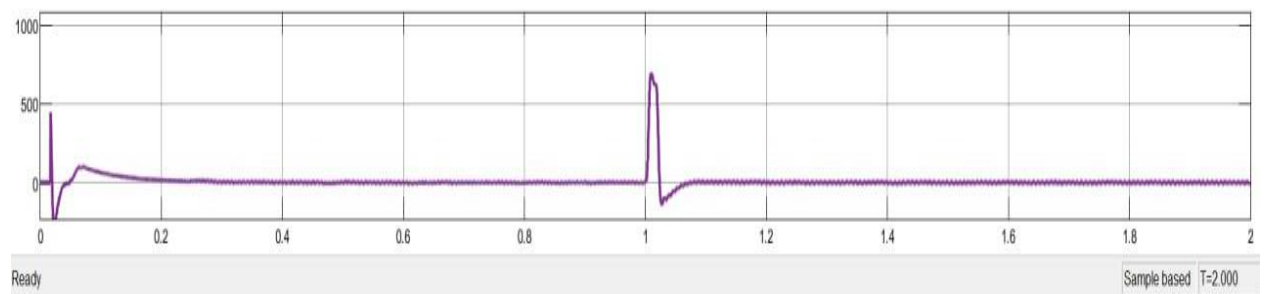
Super Capacitor Voltage Vs Time



Super Capacitor Current Vs Time



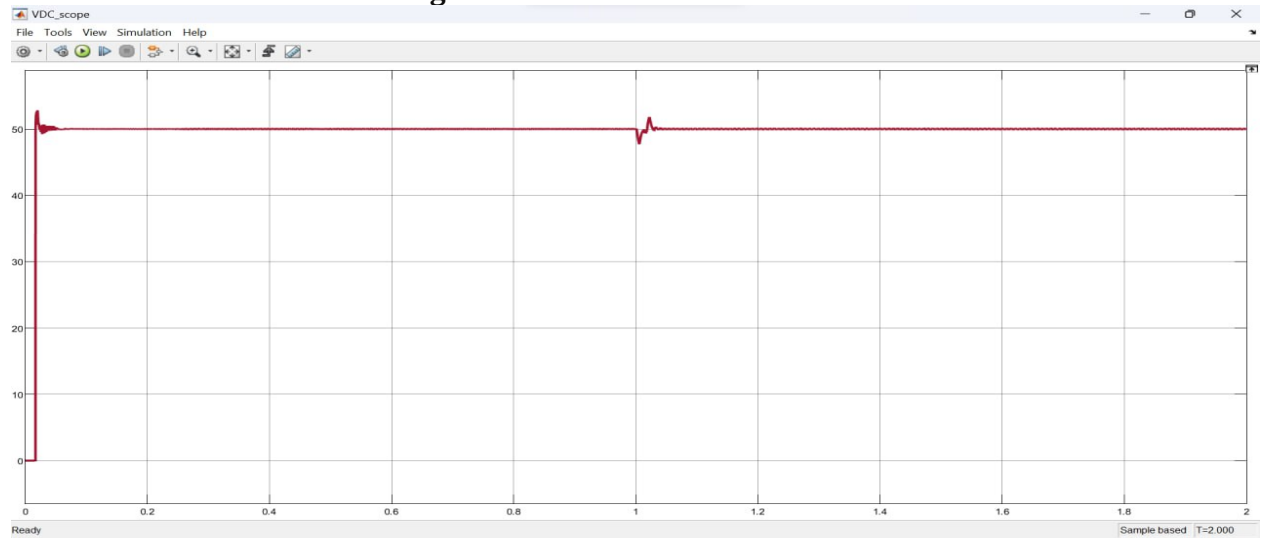
Super Capacitor Power Vs Time



Ready

Sample based T=2.000

DC-DC Boost Converter Voltage Vs Time



- Here, 25°C is always maintained
- After 1s Irradiance(amount of sunlight) slow down
- Every times after a few moments all the parameters (Voltage, Current, Power) get stabilize.

5 Design Analysis and Evaluation

5.1 Novelty:

Droop Control for Decentralized Inverter-Based AC, Islanded, and DC Microgrids represents a concerted effort to address challenges and improve the overall performance, efficiency, and adaptability of droop control strategies in the context of contemporary energy systems. The key focal points are:

1. **Advanced Control Algorithms:** Exploration and integration of sophisticated control algorithms leveraging artificial intelligence, machine learning, and advanced optimization techniques. Implementation of adaptive control strategies capable of dynamically adjusting parameters in response to varying operating conditions and system characteristics. Development of resilient droop control systems capable of swift response to disturbances, fault conditions, or alterations in microgrid configurations while maintaining stability. Research on strategies to enhance robustness and stability under challenging scenarios, contributing to the overall reliability of decentralized inverter-based microgrids.
2. **Integration of Energy Storage Systems:** Investigation into the effective integration of energy storage systems with droop control mechanisms. Optimization of coordination between inverters and energy storage, aiming to improve system performance, efficiency, and flexibility.
3. **Communication and Coordination Strategies:** Advancements in communication protocols and coordination mechanisms among decentralized inverters to optimize information exchange and ensure effective control strategies. Novel approaches to enhance the overall communication and coordination efficiency, fostering improved synchronization and control in microgrid environments.
4. **Cybersecurity Considerations:** Implementation of cybersecurity measures tailored to the unique characteristics of microgrids, safeguarding against malicious attacks and unauthorized access. Development of secure droop control strategies to mitigate potential cyber threats, ensuring the integrity and reliability of decentralized inverter-based AC, islanded, and DC microgrids. These advancements collectively contribute to the evolution of droop control in decentralized inverter-based microgrids, offering solutions that enhance adaptability, reliability, and overall performance in the rapidly evolving landscape of renewable energy and distributed power systems.

5.2 Design Considerations (PO(c))

5.2.1 Considerations to public health and safety

1. **Overvoltage and Overcurrent Protection:** Inverter controllers should incorporate overcurrent and overvoltage protection mechanisms to safeguard against faults and excessive stresses on the grid. These protections help prevent damage to equipment and reduce the risk of electrical fires.
2. **Fault Clearing:** Rapid fault clearing is essential to minimize downtime and prevent prolonged disturbances. Inverter controllers should have the capability to detect and clear faults efficiently, ensuring a quick restoration of power and reducing the impact on public health and safety.
3. **Training and Standard Operating Procedures:** Personnel responsible for the operation and maintenance of microgrid systems should be well-trained, and standard operating procedures should be established. This helps in responding effectively to emergencies, ensuring a timely and safe resolution of issues.
4. **Compliance with Standards and Regulations:** Ensure that the microgrid design and operation comply with relevant safety standards and regulations. Adhering to industry guidelines helps in mitigating risks and ensuring a high level of public safety.

5.2.2 Considerations to environment

1. **Environmental Impact:** Assess the environmental impact of the microgrid, including factors such as noise levels, electromagnetic interference, and any potential ecological effects. Strive to minimize the environmental footprint of the microgrid components and operations.
2. **Temperature and Cooling:** Inverters are sensitive to temperature variations. Consider the ambient temperature conditions and implement adequate cooling systems to ensure optimal performance and longevity of the inverters.
3. **Renewable Energy Source Variability:** Microgrids often incorporate renewable energy sources like solar and wind. These sources are variable and can introduce fluctuations in power generation. The Droop control system should be able to accommodate these variations to maintain stable operation.

5.2.3 Considerations to societal needs

1. **Reliability and Resilience:** Ensure that the microgrid can operate reliably and provide a continuous power supply to critical infrastructure, such as hospitals, emergency services, and communication systems. Implement redundancy and backup mechanisms to enhance resilience and minimize the impact of failures or disruptions.
2. **Community Engagement:** Involve the local community in the planning and decision-making processes to address specific energy needs and ensure that the microgrid aligns with community priorities. Consider educational programs to increase awareness and understanding of the microgrid system within the community.

3. **Economic Viability:** Evaluate the economic feasibility of the microgrid to ensure affordability for both the community and the individuals involved. Consider financing models that support the long-term sustainability of the microgrid while minimizing the financial burden on end-users.

5.3 Investigations (PO(d))

5.3.1 Design of Experiment

Embarking on the design and simulation of three distinctive microgrid systems with droop control independently in Simulink involves a creative journey through each design phase. We create unique models for the AC, DC, and islanded microgrids, sculpting components with Simulink blocks to breathe life into generators, inverters, and loads, each with its distinctive mathematical allure.

In the realm of the AC microgrid, we orchestrate a symphony of frequency and voltage droop controls. Through the Simulink canvas, we tune the droop coefficients to craft a harmonic balance, ensuring proportional power sharing among the microgrid's entities. The system dances to the beat of decentralized operation, each component pirouetting autonomously in response to local measurements.

In the DC microgrid realm, a canvas awaits our artistic touch. We sketch the contours of voltage droop control with Simulink blocks, adjusting coefficients to sculpt the desired voltage response. We witness a ballet of proportional power sharing, where Simulink choreographs the decentralized and distributed movements of sources and converters.

Venturing into the islanded microgrid cosmos, Simulink unveils a vibrant palette. We capture the essence of islanding operation through simulations that transition gracefully into autonomy. Simulink's brush integrates the whims of renewable sources, orchestrating a resilient performance in the face of unforeseen disturbances.

Summoning these Simulink-driven spectacles, our creations materialize on the digital stage. We run simulations with a click, observing the intricate dances unfold. Each microgrid, a unique narrative etched in Simulink's storyboard, awaits documentation and reporting. Our journey, a creative odyssey, unfolds as the microgrid sagas are captured for future inspiration.

5.3.2 Results and Analysis

Topic	AC microgrid	DC microgrid	Islanded microgrid
Active Power Vs Time	Active power increases at t=1s	Active power decreases as we shut irradiance at t=1s	Active power increases at t=1s
Frequency Vs Time	Frequency decreases as we increase active power at t=1s	Frequency unchanged	Frequency decreases as we increase active power at t=1s
Voltage Vs Time	Voltage decreases as we increase active power at t=1s	Voltage remains unchanged as we used bi-directional dc-dc converter	Voltage decreases as we increase active power at t=1s
Effect of Droop control At t=3s	Voltage and frequency stabilized	Voltage stabilized	Voltage and frequency stabilized

5.3.3 Interpretation and Conclusions on Data

5.4 Limitations of Tools (PO(e))

Computational Resources:

Limitation: Simulating complex droop control projects with a large number of components can be computationally intensive, requiring substantial processing power and memory.

Impact: Limited computational resources may lead to longer simulation times, especially for real-time or hardware-in-the-loop (HIL) simulations.

Real-Time Constraints:

Limitation: Achieving real-time performance in simulations, especially for hardware-in-the-loop setups, can be challenging due to computational delays and communication overhead.

Impact: Real-time constraints are crucial in applications where the control system interacts with physical hardware, and deviations from real-time behavior can affect simulation accuracy.

Numerical Accuracy:

Limitation: Simulink uses numerical solvers for solving differential equations, which may introduce numerical errors.

Impact: Inaccuracies can accumulate over time, affecting the precision of simulation results. Proper solver selection and settings are crucial for maintaining numerical accuracy.

Complexity of Models:

Limitation: Extremely complex models with numerous interconnected subsystems may lead to simulation challenges.

Impact: Simulink's ability to handle highly intricate models may be limited by computational resources, potentially requiring simplifications or alternate modeling approaches.

Limited Real-Time Hardware Support:

Limitation: Real-time simulation with specific hardware interfaces may have limitations in terms of supported hardware and protocols.

Impact: Integrating Simulink with real-time systems or hardware-in-the-loop setups might require additional considerations and potentially specialized hardware.

Modeling and Simulation Tool Compatibility:

Limitation: Interoperability with other modeling and simulation tools may pose challenges.

Impact: Importing/exporting models or data between Simulink and other tools might require additional efforts, and some features may not be seamlessly transferable.

Validation Challenges:

Limitation: Validating simulation results against real-world scenarios can be challenging due to uncertainties in modeling real-world complexities.

Impact: Achieving high-fidelity validation might be constrained by the accuracy of the model and limitations in representing physical phenomena.

Learning Curve:

Limitation: Simulink has a learning curve, and users may need time to become proficient in designing and simulating complex control systems.

Impact: Novice users may face challenges in efficiently utilizing Simulink's features, potentially leading to suboptimal model design or simulation setups.

5.5 Impact Assessment (PO(f))

5.5.1 Assessment of Societal and Cultural Issues

In our collective endeavor, the project emerges as a dynamic force, sculpting a nuanced impact on both cultural and societal landscapes. Infused with a commitment to inclusivity, our software solutions are crafted with cultural sensitivity at their core. The mission extends beyond mere functionality, as we actively strive to bridge digital divides, empowering communities to co-create and engage in meaningful ways.

Ethical considerations serve as our compass, ensuring that our technological innovations align harmoniously with societal values. From the preservation of cultural heritage to the cultivation of cross-cultural collaboration, our vision extends beyond the binary of code to weave a rich tapestry of diverse perspectives.

In this shared journey, technology transcends its conventional role, transforming into a potent catalyst for positive societal and cultural evolution. Our project aspires to be more than just a collection of algorithms; it aims to be a driving force for a world where diversity

not only thrives but is celebrated through every line of code and every digital interaction.

5.5.2 Assessment of Health and Safety Issues

Our project, centered on droop control implementation, holds profound implications. We prioritize energy equity, ensuring fair power distribution for inclusive access. Sustainability is a key focus, aligning with environmental responsibility and cultural values. Our commitment extends to decentralized empowerment, empowering communities and adapting seamlessly to diverse cultural landscapes.

Additionally, our project enhances community resilience through autonomous energy management and fosters cross-cultural collaboration for global sustainability. Education and awareness initiatives promote understanding, contributing to a broader cultural dialogue on sustainable practices. We harmonize with cultural values to preserve heritage and ensure technological inclusivity, bridging gaps for widespread benefit. Social acceptance is a priority, ensuring droop control seamlessly integrates into societal norms.

In essence, our project transcends technical innovation, influencing cultural and societal evolution in energy management.

5.5.3 Assessment of Legal Issues

As a team, our project on droop control involves a thorough examination of legal aspects that significantly influence our implementation strategy. Our primary focus is on ensuring compliance with regulations and industry standards in the intricate domain of energy management.

One of our main priorities is to ensure that our approach adheres to relevant laws and regulations governing energy management. This involves a meticulous examination of licensing and certification requirements to validate the legal standing and recognition of our technological solutions.

We collectively assess and manage the potential liabilities and risks associated with implementing droop control. Our goal is to identify and address any legal risks effectively, promoting responsible and legally compliant deployment of our technology.

Protecting data privacy and security is crucial, and as a team, we ensure that our data handling practices not only meet legal requirements but also uphold user privacy in line with ethical and legal expectations.

Respecting intellectual property rights is integral to our project, and we collaborate to navigate existing patents, trademarks, and copyrights while safeguarding our innovations to ensure legal compliance.

In terms of contractual agreements, we work together to draft and review contracts, ensuring legal clarity, enforceability, and fulfillment of obligations with all stakeholders involved.

Environmental compliance is also central to our approach, ensuring that our project aligns with legal requirements related to energy efficiency, emissions, and other environmental

considerations.

Adhering to public safety regulations is a collective responsibility, and we prioritize legal standards to ensure the safe implementation of droop control technologies for the well-being of the public.

Legal considerations extend to our community engagement initiatives, where we ensure compliance with local laws and regulations to align with legal requirements.

Finally, we uphold ethical principles alongside legal standards, aiming for a responsible and lawful deployment of innovative energy management technologies that positively contribute to both ethical and legal frameworks.

5.6 Sustainability Evaluation (PO(g))

As a unified team, our sustainability assessment of the droop control project is comprehensive, focusing on various dimensions to ensure lasting positive impacts. We meticulously scrutinize the environmental effects, emphasizing energy efficiency, emissions reduction, and responsible resource use. Our goal is to minimize the project's environmental footprint and champion sustainable energy practices.

Simultaneously, we delve into resource conservation, evaluating how our project optimizes energy usage and fosters efficient management strategies. This approach contributes to the conservation of natural resources, promoting sustainability for current and future generations.

Ensuring the resilience and adaptability of our droop control system is paramount. Through thorough evaluations, we confirm its capacity to withstand environmental changes and disruptions, ensuring its prolonged effectiveness.

Beyond environmental considerations, our assessment extends to social and economic sustainability. We scrutinize the project's impact on communities and economies, aiming to facilitate equitable access to sustainable energy solutions and support economic development.

Lifecycle analyses are a key component, providing insights into environmental and social impacts from production to disposal. Stakeholder engagement is actively pursued, fostering collaboration and transparency to meet the needs of diverse perspectives.

Regulatory compliance is a cornerstone, ensuring alignment with sustainability standards. Our commitment to sustainability is an ongoing process, prioritizing continuous improvement through monitoring, evaluation, and feedback mechanisms to elevate the project's sustainability profile.

5.7 Ethical Issues (PO(h))

As a cohesive team, our project on droop control is underpinned by a careful consideration of ethical issues, reflecting a commitment to responsible technological development.

Firstly, we recognize the ethical imperative of ensuring equitable access to droop control technologies. Our efforts are guided by the principle of minimizing disparities and ensuring that the benefits of our project are accessible to diverse communities. This involves a proactive approach to prevent potential exclusions or biases that may arise during implementation.

Ethical handling of data privacy is a paramount concern. Collectively, we strive to implement robust privacy measures, safeguarding sensitive information associated with droop control systems. This commitment extends throughout the project's lifecycle, demonstrating our dedication to respecting user confidentiality.

Transparency and accountability are central tenets of our ethical framework. We commit to open communication regarding the purpose, functionality, and potential impacts of droop control technologies. This transparency fosters trust and accountability within the communities affected by our project.

Community engagement is approached with ethical considerations in mind. We actively seek input from diverse communities impacted by droop control implementation, valuing their perspectives and incorporating them into decision-making processes. This ensures inclusivity and ethical representation in the development and deployment of our technologies.

Our ethical stance extends to environmental responsibility. We prioritize efforts to minimize any potential negative environmental impacts associated with droop control systems, aligning with sustainable and eco-friendly practices. This commitment reflects our dedication to ethical considerations concerning the planet's well-being.

Cultural sensitivity is another key ethical consideration. Our team recognizes the importance of understanding and respecting cultural nuances in the implementation of droop control technologies. This approach ensures that our innovations align harmoniously with diverse cultural contexts, promoting ethical practices in cross-cultural settings.

Safety and reliability are paramount ethical principles guiding our project. We strive to design and implement droop control technologies that prioritize the well-being of users and the broader community. This commitment ensures that ethical considerations are central to decisions impacting the safety and reliability of our systems.

Continuous ethical reflection is integrated into our project approach. Throughout the project lifecycle, we engage in regular assessments of the ethical implications of our decisions, innovations, and actions. This ongoing reflection ensures that our project remains aligned with evolving ethical standards and principles.

Inclusive decision-making is a key aspect of our ethical approach. We actively seek diverse perspectives within our team, fostering a collaborative environment that ensures ethical considerations are thoroughly examined from various angles. This commitment supports responsible and inclusive decision-making in the development of droop control technologies.

Finally, our ethical considerations extend to the long-term societal impact of droop control technologies. We proactively assess how our project may shape societal dynamics, aiming to contribute positively to societal well-being and align with ethical advancements in energy management.

In summary, our project on droop control is characterized by a comprehensive ethical framework that encompasses equitable access, privacy, transparency, community engagement, environmental responsibility, cultural sensitivity, safety, continuous reflection, inclusive decision-making, and considerations for the long-term societal impact of our technological advancements.

6 Reflection on Individual and Team work (PO(i))

Team communication:

Projecting at the same time made communication easier between us. We all tried to help each other in any need.

Leadership and Coordination:

During our software simulation part we had to face difficulties in finding the model of variable capacitor, and so the task of simulation of proper circuit models was then distributed among the members and coordinated on a regular basis to keep regular updates on project.

Accountability:

Reflection instills a sense of accountability as members take ownership of their actions, decisions, and outcomes.

Feedback Integration:

It provides an opportunity to integrate feedback received from team members and provides scope of mutual cooperation and professional development.

Team Dynamics:

Reflecting on team work helps in understanding team dynamics, including communication patterns, roles, and interpersonal relationships.

Being practical:

This kind of project makes us think about everything practically. How to solve any real life problem. We are learning that from this project.

6.1 Individual Contribution of Each Member

Student Id	Contribution
2006162	1.Designing & Implementing 3 types of models 2.Report Writing
2006150	1.Co-operating in the implementation process 2. Report Writing
2006154	1.Co-operating in the designing process 2. Report Writing
2006137	1.Designing & Implementing 3 types of models 2.Report Writing
2006148	1.Report writing

7 Communication to External Stakeholders (PO(j))

7.1 Executive Summary

Our project employs Simulink to implement droop control in AC, DC, and islanded microgrids, enhancing stability and efficiency. In AC microgrids, Simulink models droop control for optimal power distribution and frequency management. Similarly, in DC microgrids, Simulink regulates voltage and current levels, refining control algorithms for varied loads. For islanded microgrids, Simulink facilitates the design and simulation of droop control strategies, ensuring seamless standalone operation. Leveraging Simulink's versatility, our research aims to optimize droop control algorithms for robust performance in diverse microgrid environments, contributing to advancements in sustainable and efficient energy management systems.

7.2 YouTube Link

- https://youtu.be/Ub8iL2laX48?si=OJ34G_dn8NC_Wv5M
- <https://youtu.be/khQcHFJkYVs?si=WD0I5aFajssUI1cX>
- <https://youtu.be/KE7yp5uuuIQ?si=i1QAUi9oK8durR4z>

8 Future Work (PO(l))

Our future research on the Simulink microgrid simulation project involves optimizing control algorithms for AC, DC, and islanded microgrids. Additionally, we plan to expand the simulation to include more renewable energy sources, integrate energy storage systems, and conduct cybersecurity analyses for enhanced system resilience.

Transitioning to real-world implementation through Hardware-in-the-Loop (HIL) testing will validate simulation results, and exploring grid interconnection dynamics will provide insights into stability and support services. Lastly, we aim to analyze the impact of market dynamics on microgrid operation, optimizing for economic benefits while ensuring grid reliability.

In summary, our future work aims to refine control strategies, incorporate renewable energy, enhance resilience, validate simulations in real-time, explore grid interactions, and optimize for economic efficiency in microgrid systems.

9 References

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- <https://ietresearch.onlinelibrary.wiley.com/doi/full/10.1049/iet-rpg.2016.0500>
- <https://ieeexplore.ieee.org/abstract/document/7512558>
- <https://ieeexplore.ieee.org/abstract/document/6909049>